

29 Use of the Data Booklet is relevant to this question.

Vanadium forms a number of coloured aqueous ions.

| ion           | colour |
|---------------|--------|
| $V^{2+}(aq)$  | violet |
| $V^{3+}(aq)$  | green  |
| $VO^{2+}(aq)$ | blue   |
| $VO_2^+(aq)$  | yellow |

What are the colours of the resulting solutions if an acidic solution of  $VO_2^+(aq)$  is reacted separately with an excess of zinc metal and with an excess of tin metal?

|          | with zinc metal | with tin metal |
|----------|-----------------|----------------|
| <b>A</b> | blue            | green          |
| <b>B</b> | blue            | violet         |
| <b>C</b> | violet          | blue           |
| <b>D</b> | violet          | green          |

30 The atomic radii and the first ionisation energies of the metals iron and cobalt are very similar.